

A NONPARAMETRIC MODE ESTIMATE UNDER LTRC MODEL AND DEPENDENT DATA

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Summary: In this paper, we consider the estimation of the mode function when the data are subject to random left truncation and right censorship. Under suitable conditions, we drive almost sure (a.s.) convergence and asymptotic normality for a kernel estimator of the mode, when observed data exhibit α -mixing dependence.

1. Introduction

Let (Y, T, W) denote random variables, where Y is the interest variable called lifetime variable, with density function f and distribution function (d.f.) F . T is the random left truncation time with continuous d.f. L and W denote the random right censoring time with arbitrary d.f. G . In the random left truncated and right censored (LTRC) model one can observe (Z, T, δ) if $Z \geq T$, where $Z = \min(Y, W)$ and $\delta = I(Y \leq W)$ is the indicator of censoring status, and when $Z < T$ nothing is observed. Clearly, if Y is independent of W , then Z has d.f. $H = 1 - (1 - F)(1 - G)$. Taking into account the truncation and the censorship effects, we denote by $\{(Z_i, T_i, \delta_i), i = 1, \dots, n\}$ the actually observed sample (i.e. $Z_i \geq T_i, \forall i$) and suppose that $\mu = P[T \leq Z] > 0$.

Let

$$C(y) = P[T \leq y \leq Z | T \leq Z] = \mu^{-1} L(y) (1 - H(y^-)), \quad (1)$$

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and define their empirical estimators as

$$C_n(y) = \frac{1}{n} \sum_{i=1}^n I(T_i \leq y \leq Z_i).$$

For LTRC data, it is important to be able to obtain nonparametric estimators of the various characteristics of the distribution function, F . Tsai, Jewell and Wang (1987) gave the nonparametric maximum likelihood estimator of F itself, called the product-limit (PL) estimator, as

$$1 - \hat{F}_n(t) = \prod_{\{i: Z_i \leq t\}} \left(1 - \frac{1}{nC_n(Z_i)}\right)^{\delta_i}. \quad (2)$$

$\hat{F}_n$ reduces to the Kaplan-Meier product limit (Kaplan and Meier, 1958) estimator when $T = 0$ and to the Lynden-Bell (1971) estimator when there is no right censoring. Under independent and identically distributed (i.i.d) assumptions, the properties of $\hat{F}_n(\cdot)$ have been studied by many authors. See Gijbels and Wang (1993), Zhou (1996) and references therein. Nonparametric estimators of the density and hazard rate for $F(\cdot)$ have been studied by Sun and Zhou (1998) among others. The problem of estimating the unconditional/conditional mode of the probability density has given way to a large amount of related statistics literature. In the complete-data case (no censoring, no truncation), the mode estimator for i.i.d. random variables have been studied by many authors whom we quote Parzen (1962), Romano (1988) and Vieu (1996). For the complete dependent data case, the strong consistency of the conditional mode estimator was established under a ϕ -mixing condition by Collomb, Hardle and Hassani (1987) and their results can be applied to process forecasting. In the α -maxing case, the strong consistency over a compact set and the asymptotic normality were obtained by Ould-Saïd (1993) and Louani and Ould-Saïd (1999), respectively. In the incomplete data case, for i.i.d. random variables under random right censoring, Louani (1998) studied the asymptotic normality of the kernel estimator of the mode, while Khardani, Lemdani and Ould-Saïd (2010) established **the** some asymptotic properties for a smooth kernel estimator of the conditional mode under random censorship. These authors also studied the strong uniform consistency of the mode estimator for censored time series (Khardani, Lemdani and Ould-Saïd, 2012). Under random left truncation (RLT model), in the i.i.d. case, Ould-Saïd and Tatachak (2009a) studied the nonparametric estimation of the simple mode under left-truncation model. They also studied the strong consistency rate for the kernel mode estimator under strong mixing hypothesis and left truncation Ould-Saïd and Tatachak (2009b).

The aim of this paper is to establish a strong uniform convergence rate and asymptotic normality for a kernel mode estimator with randomly left truncated and right censored data under α -mixing condition.

We begin by recalling the definition of the strong property, for this we introduce the following notations.

Let $\{U_i, i \geq 1\}$ denote a sequence of random variables and $\mathcal{F}_i^k(U)$ denote the σ -algebra of events generated by $\{U_j, i \leq j \leq k\}$.

Given a positive integer n , set:

$$\alpha(n) = \sup \left\{ |P(A \cap B) - P(A)P(B)| : A \in \mathcal{F}_1^k(U), B \in \mathcal{F}_{k+n}^\infty(U), k \in \mathbb{N}^* \right\}. \quad (3)$$

The sequence is said to be α -mixing if the mixing coefficient $\alpha(n) \rightarrow 0$ as $n \rightarrow \infty$.

The α -mixing condition is the weakest among mixing conditions known in the literature. Many stochastic processes satisfy the α -mixing condition; the ARMA processes are geometrically strongly mixing, i.e., there exists $\rho \in (0, 1)$ such that, $\alpha(n) = O(\rho^{-n})$ (see, e.g., Jones, 1978).

The rest of the paper is organized as follows: In section 2, we define a new kernel mode estimator in LTRC model with some notations. Assumptions and main results are given in section 3. Finally, the proofs of the main results are relegated to section 4 with some auxiliary results.

2. Definition of the estimator

For any distribution function (d.f.) K , let $a_K = \inf\{x : K(x) > 0\}$ and $b_K = \sup\{x : K(x) < 1\}$ denote the left and right end points of its support. In the current model, as discussed by Gijbels and Wang (1993) and Zhou (1996), we assume that

$$a_L \leq a_H, \quad b_L \leq b_H \quad \text{and} \quad \int_{a_H}^{\infty} \frac{dF(x)}{L^2(x)} < \infty. \quad (4)$$

We denote by a superscript $(*)$ any d.f. that is associated to the observed sample.

Now, let $H_1^*(\cdot)$ be defined as

$$H_1^*(z) = P[Z \leq z, \delta = 1 \mid T \leq Z] = \mu^{-1} \int_{-\infty}^z L(y) (1 - G(y^-)) dF(y) \quad (5)$$

where $H_1^*(z)$ is consistently estimated as

$$H_{1n}^*(z) = \frac{1}{n} \sum_{i=1}^n I(Z_i \leq z, \delta_i = 1).$$

According to (1) and by replacing F , L and G by their respective nonparametric estimators, we can consider the estimator of μ

$$\mu_n = \frac{L_n(y) (1 - \hat{F}_n(y^-)) (1 - G_n(y))}{C_n(y)}.$$

For all y when $C_n(y) > 0$, and $\hat{F}_n(y^-)$ denotes the left limit of $\hat{F}_n$ at y . Here, L_n is the product-limit estimator Lynden-Bell for L

$$L_n(y) = \prod_{\{i: T_i > y\}} \left(1 - \frac{1}{nC_n(T_i)}\right), \quad (6)$$

and G_n is the product-limit estimator Kaplan and Meier for G

$$G_n(y) = 1 - \prod_{\{i: Y_i \leq y\}} \left(1 - \frac{1 - \delta_i}{n - (i - 1)}\right). \quad (7)$$

For the complete data, it is well known that the kernel estimator of the mode θ is defined as the random variable θ_n

$$\theta_n = \arg \max_{t \in \mathbb{R}} f_n(t) \quad (8)$$

where

$$f_n(t) = \frac{1}{nh_n} \sum_{i=1}^n K\left(\frac{t-Y_i}{h_n}\right)$$

where, K is a probability density function (so-called kernel function) and (h_n) is a sequence of positive real numbers (so-called bandwidth) which goes to zero as n goes to infinity.

Under *RTL* model, Ould-Saïd and Tatachak (2009a) define a new estimator $\hat{f}_n(\cdot)$ based on the n actually observed pair (Y_i, T_i) given as

$$\hat{f}_n(t) = \frac{\mu_n}{nh_n} \sum_{i=1}^n \frac{1}{L_n(Y_i)} K\left(\frac{t-Y_i}{h_n}\right).$$

In the right censoring model, Khardani et al. (2012) define an estimator $\hat{f}_n(\cdot)$ based on the kernel smoothing technique, given by

$$\hat{f}_n(t) = \frac{1}{nh_n} \sum_{i=1}^n \frac{\delta_i}{\bar{G}_n(Y_i)} K\left(\frac{t-Y_i}{h_n}\right)$$

where $\bar{G} = 1 - G$.

Now, we define a new estimator $\hat{f}_n(\cdot)$ in *LTRC* model based on the kernel smoothing technique, given as

$$\hat{f}_n(t) = \frac{\mu_n}{nh_n} \sum_{i=1}^n \frac{\delta_i}{L_n(Z_i) \bar{G}_n(Z_i)} K\left(\frac{t-Z_i}{h_n}\right) \quad (9)$$

Then a natural estimator of θ is

$$\hat{\theta}_n = \arg \max_{t \in [a_H, b_H]} \hat{f}_n(t). \quad (10)$$

It is noticed that the estimator $\hat{\theta}_n$ is not necessarily unique and our results are valid for any chosen value satisfying (10). We point out that our choice can be specified by taking

$$\hat{\theta}_n = \inf \left\{ t \in [a_H, b_H] \text{ such that } \hat{f}_n(t) = \max_{y \in [a_H, b_H]} \hat{f}_n(y) \right\}.$$

Suppose that K is chosen to be twice differentiable, we get the derivative of $\hat{f}_n(\cdot)$ as

$$\hat{f}_n^{(j)}(t) = \frac{\mu_n}{nh_n^{j+1}} \sum_{i=1}^n \frac{\delta_i}{L_n(Z_i) \bar{G}_n(Z_i)} K^{(j)}\left(\frac{t-Z_i}{h_n}\right) \quad \text{for } j \geq 1.$$

Adapting (9), we define

$$\tilde{f}_n(t) = \frac{\mu}{nh_n} \sum_{i=1}^n \frac{\delta_i}{L(Z_i) \bar{G}(Z_i)} K\left(\frac{t-Z_i}{h_n}\right).$$

3. Assumptions and main results

In what follows, we suppose that $a_L \leq a_H$, $b_L \leq b_H$ and $D = [a, b]$ be a compact set such that $D \subset D_0 = \{t : t \in [a_H, b_H]\}$.

Now, some assumptions needed to state our results, are introduced and gathered below for easy reference.

A1: The bandwidth h_n satisfies:

- (i) $\sqrt{\frac{\log \log n}{n}} = O(h_n^3)$;
- (ii) $\lim_{n \rightarrow \infty} \frac{nh_n^{(v+4)/(v-4)}}{(\log n)^{\frac{v+1}{v-4}} (\log \log n)^{\frac{6}{v-4}}} \rightarrow \infty$ as $n \rightarrow \infty$ (where v is as in **A3**).

A2: The density function K satisfies:

- (i) K is three times differentiable with compact support on $\mathbb{R}$;
- (ii) K and $K^{(2)}$ are Lipschitz continuous functions;
- (iii) $\int_{\mathbb{R}} tK(t) dt = 0$, $\int_{\mathbb{R}} t^2 K(t) dt < \infty$.

A3: $(Y_i)_{i \geq 1}$ is a stationary α -mixing sequence of rv's, with coefficient $\alpha(n) = O(n^{-v})$ for some $v > 4$.

A4: The probability density $f(\cdot)$ is differentiable up to order 3 and $f^{(2)}(\cdot)$ does not vanish.

A5: The joint density $f_{1j}(\cdot, \cdot)$ of (Y_1, Y_j) exists and satisfies for any constant $M > 0$

$$\sup_{r,t} |f_{1j}(r,t) - f(r)f(t)| \leq M < \infty$$

A6: The mode θ satisfies the following property: for any $\varepsilon > 0$ and t , there exists a $\eta > 0$ such that $|\theta - t| \geq \varepsilon$ implies $|f(\theta) - f(t)| \geq \eta$.

The following assumptions intervene in the asymptotic normality results

B1: The bandwidth (h_n) satisfies $nh_n^7 \rightarrow 0$ as $n \rightarrow \infty$.

B2: Let (M_n) and (N_n) be subsequences of (n) tending to infinity such that:

- (i) $M_n + N_n \leq n$; $\frac{r_n M_n}{n} \rightarrow 1$ and $\frac{r_n N_n}{n} \rightarrow 0$;
- (ii) $M_n (nh_n)^{-1/2} \rightarrow 0$;
- (iii) $r_n \alpha(N_n) \rightarrow 0$, as $n \rightarrow \infty$ where (r_n) is the largest positive integer for which $r_n (M_n + N_n) \leq n$.

Remark 3.1. (Discussion of the assumptions)

Assumption **A1**(i) is needed in the behavior study of the difference between $\hat{f}_n(\cdot)$ and $\tilde{f}_n(t)$, while **A1**(ii) gives a condition for the bandwidth which allows to estimate the covariance term. Assumptions

A2 and **A4** are common in nonparametric estimation. Assumption **A3** specifies the considered models as well as the rate for the mixing coefficient. Assumption **A5** is needed for covariance calculus. Assumption **A6** stipulates the uniform uniqueness of the mode function. Assumption **B1** is used to prove the asymptotic normality result as well as the convergence of $\hat{f}_n^{(2)}(\cdot)$. Finally, assumption **B2** deals with real sequences that take part in establishing our results (see Louani and Ould-Saïd, 1999).

Remark 3.2. Assumption **A2(i)** implies that the kernels K , $K^{(1)}$ and $K^{(2)}$ are bounded by constants M , M' and M'' respectively.

3.1. Consistency

In this subsection, we prove the consistency of our estimator and give a rate of convergence.

Proposition 3.1. Under assumptions **A1-A5**, we have

$$\sup_{t \in D} |\hat{f}_n(t) - f(t)| = O \left\{ \max \left(\left(\frac{\log n}{nh_n} \right)^{1/2}, h_n^2 \right) \right\} \text{ a.s. as } n \rightarrow \infty.$$

Theorem 3.1. Under the assumptions of **Proposition 3.1** and **A6**, we have

$$\hat{\theta}_n - \theta = O \left\{ \max \left(\left(\frac{\log n}{nh_n} \right)^{1/4}, h_n \right) \right\} \text{ a.s. as } n \rightarrow \infty.$$

3.2. Asymptotic normality

Now, suppose that the density function $f(\cdot)$ is unimodal at θ . Under **A6**, we have

$$f^{(1)}(\theta) = 0 \text{ and } f^{(2)}(\theta) < 0.$$

Similarly, we have

$$\hat{f}_n^{(1)}(\hat{\theta}_n) = 0 \text{ and } \hat{f}_n^{(2)}(\hat{\theta}_n) < 0.$$

Using a Taylor expansion, we get

$$0 = \hat{f}_n^{(1)}(\hat{\theta}_n) = \hat{f}_n^{(1)}(\theta) + (\hat{\theta}_n - \theta) \hat{f}_n^{(2)}(\bar{\theta}_n)$$

where $\bar{\theta}_n$ is between $\hat{\theta}_n$ and θ , which gives that

$$\hat{\theta}_n - \theta = - \frac{\hat{f}_n^{(1)}(\theta)}{\hat{f}_n^{(2)}(\bar{\theta}_n)} \quad (11)$$

if the denominator does not vanish.

To establish the asymptotic normality, we show that the numerator in (11), suitably normalised, is asymptotically normally distributed and that the denominator converges in probability to $f^{(2)}(\theta)$. The result is given in the following theorem:

Theorem 3.2. Assume that assumptions **A1(i)**, **A2-A5** and **B1-B2** hold, then we have

$$\sqrt{nh_n^3}(\hat{\theta}_n - \theta) \xrightarrow{D} N(0, \sigma^2) \quad (12)$$

where $\xrightarrow{D}$ denotes the convergence in distribution and

$$\sigma^2 = \frac{\mu f(\theta)}{\bar{G}(\theta) L(\theta) [f^{(2)}(\theta)]^2} \int_{\mathbb{R}} [K^{(1)}(r)]^2 dr.$$

4. Auxiliary results and proofs

Some auxiliary results and notations are needed to be proved. The first Lemma deals with the behavior of the difference between $E[\tilde{f}_n(t)]$ and $f(t)$.

Lemma 4.1. Under assumptions **A2** and **A4**, we have

$$\sup_{t \in D} |E[\tilde{f}_n(t)] - f(t)| = O(h_n^2)$$

Proof. Indeed, using a change of variable and a Taylor expansion, we have

$$\begin{aligned} E[\tilde{f}_n(t)] - f(t) &= \frac{1}{h_n} E \left[\frac{\mu \delta_1}{L(Z_1) \bar{G}(Z_1)} K \left(\frac{t - Z_1}{h_n} \right) \right] - f(t) \\ &= \frac{1}{h_n} \int \frac{\mu}{L(y) \bar{G}(y)} K \left(\frac{t - y}{h_n} \right) dH_1^*(y) - f(t) \\ &= \frac{1}{h_n} \int K \left(\frac{t - y}{h_n} \right) f(y) dy - f(t) \\ &= \int K(s) \frac{(hs)^2}{2} f^{(2)}(\zeta) ds \end{aligned}$$

with ζ is between $t - sh$ and t .

Thus

$$|E[\tilde{f}_n(t)] - f(t)| \leq \frac{h^2}{2} \sup_{t \in D} |f^{(2)}(t)| \int s^2 K(s) ds. \quad (13)$$

Under the given condition, the result holds. $\square$

The second Lemma deals the with behavior of the difference between $\hat{f}_n(t)$ and $\tilde{f}_n(t)$.

Lemma 4.2. Under assumptions **A1(i)**, **A2** and **A4**, for n large enough we have

$$\sup_{t \in D} |\hat{f}_n(t) - \tilde{f}_n(t)| = O(h_n^2).$$

Proof. We have the following decomposition

$$\begin{aligned}
|\hat{f}_n(t) - \tilde{f}_n(t)| &\leq \frac{1}{nh_n} \sum_{i=1}^n \delta_i K\left(\frac{t - Z_i}{h_n}\right) \left| \frac{(1 - \hat{F}_n(Z_i))}{C_n(Z_i)} - \frac{(1 - F(Z_i))}{C(Z_i)} \right| \\
&\leq \underbrace{\frac{M}{h_n} \sup_{t \in D} \left| \frac{(1 - \hat{F}_n(t))}{C_n(t)} - \frac{(1 - F(t))}{C(t)} \right|}_{I} \frac{1}{n} \sum_{i=1}^n \delta_i \\
&\leq \frac{M}{h_n} \left[\sup_{t \in D} \left| \frac{\hat{F}_n(t) - F(t)}{C_n(t)} \right| + \sup_{t \in D} \left| \frac{1 - F(t)}{C_n(t)C(t)} (C_n(t) - C(t)) \right| \right] \frac{1}{n} \sum_{i=1}^n \delta_i \\
&\leq \frac{M}{h_n} \left[\underbrace{\frac{1}{C_n(a)} \sup_{t \in D} |\hat{F}_n(t) - F(t)|}_I + \underbrace{\frac{1 - F(a)}{C_n(a)C(a)} \sup_{t \in D} |C_n(t) - C(t)|}_J \right] \frac{1}{n} \sum_{i=1}^n \delta_i.
\end{aligned}$$

Since $C(a) > 0$, $I = O\left(\sqrt{\frac{\log \log n}{n}}\right)$ (see Lemma 2.1 in Chen and Dai, 2003) and $J = O\left(\sqrt{\frac{\log \log n}{n}}\right)$ (see Theorem 3.2 in Cai and Roussas, 1992).

In conjunction with the SLLN on the censoring law (see Theorem 3.2 in Cai and Roussas, 1992), we have

$$\sup_{t \in D} |\hat{f}_n(t) - \tilde{f}_n(t)| \leq \frac{c}{h_n} \sqrt{\frac{\log \log n}{n}} \text{ a.s.}$$

where c is a positive constant. With assumption **A1**(i) permits concluding the proof. $\square$

Lemma 4.3. Under assumptions **A1-A5**, we have

$$\sup_{t \in D} |\tilde{f}_n(t) - E[\tilde{f}_n(t)]| = O\left(\sqrt{\frac{\log n}{nh_n}}\right)$$

Proof. We use covering set techniques. Indeed, since D is a compact set, it can be covered by a finite number d_n of intervals centered at t_j of length λ_n such that $\lambda_n = (n^{-1}h_n^3)^{\frac{1}{2}}$. As D is bounded, there exists a constant A such that $d_n\lambda_n \leq A$. Now we put $J(t) = \arg \min_{j=1, \dots, d_n} |t - t_j|$.

Thus we have the following decomposition

$$\begin{aligned}
\sup_{t \in D} |\tilde{f}_n(t) - E[\tilde{f}_n(t)]| &\leq \max_{j=1, \dots, d_n} \sup_{t \in D} |\tilde{f}_n(t) - \tilde{f}_n(t_{J(t)})| \\
&\quad + \max_{j=1, \dots, d_n} |\tilde{f}_n(t_{J(t)}) - E[\tilde{f}_n(t_{J(t)})]| \\
&\quad + \max_{j=1, \dots, d_n} \sup_{t \in D} |E[\tilde{f}_n(t_{J(t)})] - E[\tilde{f}_n(t)]| \\
&= I_{1n} + I_{2n} + I_{3n}.
\end{aligned} \tag{14}$$

Firstly, concerning I_{1n} and I_{3n} , with assumption **A2**, we have

$$I_{1n} \leq \frac{cM}{L(a)\overline{G}(b)} \lambda_n h_n^{-2}.$$

The condition upon λ_n , we get

$$I_{1n} = O\left(\frac{1}{\sqrt{nh_n}}\right) \text{ a.s. as } n \rightarrow \infty. \quad (15)$$

As to I_{2n} , for all $\varepsilon > 0$, we have

$$P\left[\max_{j=1,\dots,d_n} |\tilde{f}_n(t_{J(t)}) - E[\tilde{f}_n(t_{J(t)})]| > \varepsilon\right] \leq d_n P[|\tilde{f}_n(t_{J(t)}) - E[\tilde{f}_n(t_{J(t)})]| > \varepsilon]. \quad (16)$$

Now, for any $i \geq 1$, we put

$$\Delta_i(t) = \frac{\mu}{nh_n} \left[\frac{\delta_i}{L(Z_i)\overline{G}(Z_i)} K\left(\frac{t-Z_i}{h_n}\right) - E\left[\frac{\delta_i}{L(Z_i)\overline{G}(Z_i)} K\left(\frac{t-Z_i}{h_n}\right)\right] \right]. \quad (17)$$

Under **A2**, the rv's $U_i(t_{J(t)}) := nh_n \Delta_i(t_{J(t)})$ are centered and bounded by $\frac{2\mu}{L(a)\overline{G}(b)} M < \infty$. The use of the well-known Fuk-Nagaev's inequality (see Rio, 2000, p. 87, 6.19b), allows us to get, for all $\varepsilon > 0$ and $r > 1$

$$\begin{aligned} & P\left[\max_{j=1,\dots,d_n} \left|\sum_{i=1}^n \Delta_i(t_{J(t)})\right| > \varepsilon\right] \\ &= P\left[\max_{j=1,\dots,d_n} \left|\sum_{i=1}^n U_i(t_{J(t)})\right| > \varepsilon nh_n\right] \\ &\leq A\lambda_n^{-1} \left[\frac{n}{r} \left(\frac{2r}{nh_n\varepsilon}\right)^{r+1} + \left(1 + \frac{n^2 h_n^2 \varepsilon^2}{r S_n}\right)^{-r/2} \right] \\ &= Q_{1n} + Q_{2n} \end{aligned}$$

where

$$S_n = \sum_{i=1}^n \sum_{j=1}^n |\text{cov}(U_i, U_j)|. \quad (18)$$

Now, we have to study the asymptotic behavior of (18). First, we calculate

$$S_n = n\text{Var}(U_1) + \sum_{i \neq j} |\text{cov}(U_i, U_j)| = n\text{Var}(U_1) + S_n^*. \quad (19)$$

By using **A2**, the conditional expectation properties and a change of variables, we get

$$\begin{aligned} n\text{Var}(U_1) &\leq nE\left[\frac{\mu^2 \delta_1}{L^2(Z_1)\overline{G}^2(Z_1)} K^2\left(\frac{t_{J(t)} - Z_1}{h_n}\right)\right] \\ &\leq \frac{nh_n \mu}{L(a)\overline{G}(b)} \int_{\mathbb{R}} K^2(s) f(t_{J(t)} - sh_n) ds \\ &= O(nh_n). \end{aligned} \quad (20)$$

Then, from (18), using again the conditional expectation, we get under **A3**

$$\begin{aligned}
 \text{cov}(U_i, U_j) &= E(U_i U_j) \\
 &= \int \int K\left(\frac{t_{J(t)} - r}{h_n}\right) K\left(\frac{t_{J(t)} - s}{h_n}\right) \\
 &\quad \times f_{1j}(r, s) dr ds \\
 &\quad - \int K\left(\frac{t_{J(t)} - r}{h_n}\right) f(r) dr \\
 &\quad \times \int K\left(\frac{t_{J(t)} - s}{h_n}\right) f(s) ds.
 \end{aligned}$$

Then

$$\begin{aligned}
 |\text{cov}(U_i, U_j)| &\leq h_n^2 \int \int K(m_1) K(m_2) \\
 &\quad \times |f_{1,j}(t_{J(t)} - h_n m_1, t_{J(t)} - h_n m_2) \\
 &\quad - f(t_{J(t)} - h_n m_1) f(t_{J(t)} - h_n m_2)| dm_1 dm_2.
 \end{aligned}$$

Using assumption **A5**, it follows that

$$|\text{cov}(U_i, U_j)| = O(h_n^2). \quad (21)$$

Now, following Masry (1986) we define the sets

$$E_1 = \{(i, j) \text{ such that } 1 \leq |i - j| \leq \varphi_n\} \text{ and } E_2 = \{(i, j) \text{ such that } \varphi_n + 1 \leq |i - j| \leq n - 1\}$$

where $\varphi_n \rightarrow \infty$ as $n \rightarrow \infty$, we can write

$$S_n^* = \mathcal{F}_{1,n} + \mathcal{F}_{2,n}$$

where $\mathcal{F}_{1,n}$ and $\mathcal{F}_{2,n}$ are the sums of covariances over E_1 and E_2 respectively. First, by applying the upper bound (21) we get

$$\mathcal{F}_{1,n} = O(nh_n^2 \varphi_n).$$

For the second term, we use the modified Davydov covariance inequality for mixing processes (see Rio, 2000, formula 1.12a, page 10),

$$\forall i \neq j, \quad |\text{cov}(U_i, U_j)| \leq c\alpha(|i - j|). \quad (22)$$

Then, we get for assumption **A3**

$$\mathcal{F}_{2,n} = O(n\varphi_n^{1-\nu}).$$

Choosing $\varphi_n = h_n^{-2/\nu}$ permits us to get

$$S_n^* = O\left(nh_n^{(\nu-1)\frac{2}{\nu}}\right). \quad (23)$$

Finally, under **A3**, combined to (20) and (23) yields

$$S_n = O(nh_n). \quad (24)$$

Therefore, by putting

$$\varepsilon = \varepsilon_0 \sqrt{\frac{\log n}{nh_n}} \quad ; \quad r = c \log n (\log \log n)^{1/\nu} \quad (25)$$

we get

$$Q_{1n} = O\left(\lambda_n^{-1} n^{(1-\nu)/2} h_n^{-(\nu+1)/2} (\log n)^{(\nu-1)/2} (\log \log n) \varepsilon_0^{-(\nu+1)}\right).$$

Under **A1(ii)**, yields $Q_{1n} = O\left(n^{-1} \log^{-1} n (\log \log^{-2} n)\right)$, which is the general term of a convergent Bertrand's series. Using a Taylor expansion of $\log(x+1)$ and (25), we have

$$\begin{aligned} Q_{2n} &= A_2 n^{1/2} h_n^{-3/2} \exp\left[-\frac{r}{2} \log\left(1 + \frac{nh_n \varepsilon_0^2 \log n}{r S_n}\right)\right] \\ &\leq c h_n^{-3/2} n^{\frac{1-\varepsilon_0^2}{2}}. \end{aligned} \quad (26)$$

In the same way, we can choose ε_0 such that Q_{2n} is the general term of convergent series. Thus, $\sum_{n \geq 1} (Q_{1n} + Q_{2n}) < \infty$, and the result is then a direct application of Borel-Cantelli's Lemma. $\square$

Proof of Proposition 3.1. Using the triangle inequality, we have

$$\sup_{t \in D} |\hat{f}_n(t) - f(t)| \leq \sup_{t \in D} |\hat{f}_n(t) - \tilde{f}_n(t)| + \sup_{t \in D} |\tilde{f}_n(t) - E[\tilde{f}_n(t)]| + \sup_{t \in D} |E[\tilde{f}_n(t)] - f(t)|$$

Then Lemmas **4.1**, **4.2** and **4.3** give the result. $\square$

Proof of Theorem 3.1. Standard argument gives us

$$\begin{aligned} |f(\hat{\theta}_n) - f(\theta)| &\leq |f(\hat{\theta}_n) - \hat{f}_n(\hat{\theta}_n)| + |\hat{f}_n(\hat{\theta}_n) - f(\theta)| \\ &\leq \sup_{t \in D} |\hat{f}_n(t) - f(t)| + |\hat{f}_n(\hat{\theta}_n) - f(\theta)| \\ &\leq 2 \sup_{t \in D} |\hat{f}_n(t) - f(t)|. \end{aligned} \quad (27)$$

The *a.s.* uniform consistency of $\hat{\theta}_n$ follows then immediately from **Proposition 3.1** and **A6**. Now a Taylor expansion gives

$$f(\hat{\theta}_n) - f(\theta) = \frac{1}{2} (\hat{\theta}_n - \theta)^2 f^{(2)}(\hat{\theta}_n^*)$$

where $\hat{\theta}_n^*$ is between $\hat{\theta}_n$ and θ . Then by (27) and **A4**, we have

$$|\hat{\theta}_n - \theta| \leq 2 \sqrt{\frac{\sup_{t \in D} |\hat{f}_n(t) - f(t)|}{|f^{(2)}(\hat{\theta}_n^*)|}}.$$

The result is then a direct consequence of **Proposition 3.1**. $\square$

Proof of Theorem 3.2. From (11) we have the following decomposition

$$\begin{aligned}
 \sqrt{nh_n^3}(\hat{\theta}_n - \theta) &= \sqrt{nh_n^3} \frac{\hat{f}_n^{(1)}(\theta) - \tilde{f}_n^{(1)}(\theta)}{\hat{f}_n^{(2)}(\bar{\theta}_n)} \\
 &\quad + \sqrt{nh_n^3} \frac{\tilde{f}_n^{(1)}(\theta) - E[\tilde{f}_n^{(1)}(\theta)]}{\hat{f}_n^{(2)}(\bar{\theta}_n)} \\
 &\quad + \sqrt{nh_n^3} \frac{E[\tilde{f}_n^{(1)}(\theta)]}{\hat{f}_n^{(2)}(\bar{\theta}_n)} \\
 &= \frac{J_1 + J_2 + J_3}{\hat{f}_n^{(2)}(\bar{\theta}_n)}. \tag{28}
 \end{aligned}$$

To prove the results, we establish that J_1 and J_3 are negligible and J_2 is asymptotically normal and $\hat{f}_n^{(2)}(\bar{\theta}_n) \rightarrow f^{(2)}(\theta)$. $\square$

Lemma 4.4. Under assumptions **A1(i)**, **A2-A4** and **B1**, we have

$$J_1 \rightarrow 0 \text{ a.s. as } n \rightarrow \infty.$$

Proof. We have

$$\begin{aligned}
 |J_1| &= \sqrt{nh_n^3} \left| \frac{1}{nh_n^2} \sum_{i=1}^n \delta_i K^{(1)}\left(\frac{\theta - Z_i}{h_n}\right) \left(\frac{(1 - \hat{F}_n(Z_i))}{C_n(Z_i)} - \frac{(1 - F(Z_i))}{C(Z_i)} \right) \right| \\
 &\leq \sqrt{nh_n} \left[\underbrace{\frac{1}{C_n(a)} \sup_{t \in D} |\hat{F}_n(t) - F(t)|}_I + \underbrace{\frac{1 - F(a)}{C_n(a)C(a)} \sup_{t \in D} |C_n(t) - C(t)|}_J \right] \frac{1}{nh_n} \sum_{i=1}^n \delta_i \left| K^{(1)}\left(\frac{\theta - Z_i}{h_n}\right) \right|.
 \end{aligned}$$

Then for any $\varepsilon > 0$, under **A2**, we have

$$P \left[\frac{1}{nh_n} \sum_{i=1}^n \left| K^{(1)}\left(\frac{\theta - Z_i}{h_n}\right) \right| > \varepsilon \right] \leq \frac{1}{nh_n \varepsilon} \sum_{i=1}^n E \left(\left| K^{(1)}\left(\frac{\theta - Z_i}{h_n}\right) \right| \right) \rightarrow \frac{f(\theta)}{\mu \varepsilon} \int_{\mathbb{R}} |K^{(1)}(r)| dr.$$

Since $C(a) > 0$, $I = O\left(\sqrt{\frac{\log \log n}{n}}\right)$ (see Lemma 2.1 in Chen and Dai, 2003) and $J = O\left(\sqrt{\frac{\log \log n}{n}}\right)$ (see Theorem 3.2 in Cai and Roussas, 1992), we have $|J_1| = O(\sqrt{h_n \log \log n})$, which goes to zero under **A1(i)** and **B1**. $\square$

Lemma 4.5. Under assumptions **A2**, **A4** and **B1**, we have

$$J_3 \rightarrow 0 \text{ a.s. as } n \rightarrow \infty.$$

Proof. Integrating by part and a change of variables, we have

$$\begin{aligned} E \left[\tilde{f}_n^{(1)}(\theta) \right] &= \frac{1}{h_n^2} E \left[\frac{\mu \delta_1}{L(Z_1) \bar{G}(Z_1)} K^{(1)} \left(\frac{\theta - Z_1}{h_n} \right) \right] \\ &= \frac{1}{h_n^2} \int_{\mathbb{R}} K^{(1)} \left(\frac{\theta - v}{h_n} \right) f(v) dv \\ &= \int_{\mathbb{R}} K(r) f^{(1)}(\theta - rh_n) dr. \end{aligned}$$

By a Taylor expansion of $f^{(1)}(\cdot)$ around θ , **A2** and the definition of the mode, we get

$$\sqrt{nh_n^3} E \left[\tilde{f}_n^{(1)}(\theta) \right] = \sqrt{nh_n^7} \int_{\mathbb{R}} r^2 K(r) f^{(3)}(\theta^*) dr$$

where θ^* is between θ and $\theta - rh_n$. By **A2**, **A4** and **B1**, we conclude that $J_3 \rightarrow 0$ a.s. as $n \rightarrow \infty$. $\square$

Lemma 4.6. Under assumptions **A2-A5** and **B2** we have

$$J_2 \xrightarrow{D} N(0, \sigma^2).$$

Proof. In order to establish the asymptotic normality for sums of dependent rv's, we use Doob's small-block and large-block technique (see Doob, 1953, pp. 228-232). Partition the set $\{1, 2, \dots, n\}$ into $2r_n + 1$ subsets with large blocks of size M_n and small blocks of size N_n , where (M_n) , (N_n) and (r_n) are sequences of integer numbers described in assumption **B2**.

Put

$$\begin{aligned} \Gamma_i &= \frac{\mu}{\sqrt{h_n}} \left\{ \frac{\delta_i}{L(Z_i) \bar{G}(Z_i)} K^{(1)} \left(\frac{\theta - Z_i}{h_n} \right) \right. \\ &\quad \left. - E \left[\frac{\delta_i}{L(Z_i) \bar{G}(Z_i)} K^{(1)} \left(\frac{\theta - Z_i}{h_n} \right) \right] \right\}. \end{aligned}$$

Then

$$J_2 = \frac{1}{\sqrt{n}} \sum_{i=1}^n \Gamma_i = \frac{1}{\sqrt{n}} [T_{1,n} + T_{2,n} + T_{3,n}]$$

where

$$T_{1,n} = \sum_{m=1}^{r_n} L_m, \quad T_{2,n} = \sum_{m=1}^{r_n} L'_m \text{ and } T_{3,n} = \sum_{i=(M_n+N_n)r_n+1}^n \Gamma_i.$$

Also set

$$L_m = \sum_{i=(m-1)(M_n+N_n)+1}^{(m-1)(M_n+N_n)+M_n} \Gamma_i \text{ and } L'_m = \sum_{i=(m-1)(M_n+N_n)+M_n+1}^{m(M_n+N_n)} \Gamma_i.$$

We first show that $\frac{1}{n} \left(E \left[T_{2,n}^2 \right] + E \left[T_{3,n}^2 \right] \right) \rightarrow 0$, as $n \rightarrow \infty$. Next, we state that $\frac{1}{\sqrt{n}} (T_{1,n})$ converges in distribution to a normal variable with 0 mean and a variance given explicitly.

Lemma 4.7. Under the assumptions of **Lemma 4.6**, we have

$$\frac{1}{n} (E [T_{2,n}^2] + E [T_{3,n}^2]) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Proof. Obviously

$$\begin{aligned} \frac{1}{n} E [T_{2,n}^2] &= \frac{1}{n} \sum_{m=1}^{r_n} \sum_{i=(m-1)(M_n+N_n)+M_n+1}^{m(M_n+N_n)} E [\Gamma_i^2] \\ &\quad + \frac{2}{n} \sum_{m=1}^{r_n} \sum_{(m-1)(M_n+N_n)+M_n+1 \leq i < j \leq m(M_n+N_n)} cov(\Gamma_i, \Gamma_j) \\ &\quad + \frac{2}{n} \sum_{1 \leq i < j \leq r_n} cov(L'_i, L'_j) \\ &= \Lambda_{1n} + \Lambda_{2n} + \Lambda_{3n}. \end{aligned}$$

First, we have

$$\begin{aligned} E [\Gamma_i^2] &= \frac{1}{h_n} E \left[\frac{\mu^2 \delta_i}{L^2(Z_i) \bar{G}^2(Z_i)} \left(K^{(1)} \right)^2 \left(\frac{\theta - Z_i}{h_n} \right) \right] \\ &\quad - \frac{1}{h_n} \left\{ E \left[\frac{\mu \delta_i}{L(Z_i) \bar{G}(Z_i)} K^{(1)} \left(\frac{\theta - Z_i}{h_n} \right) \right] \right\}^2 \\ &= \int_{\mathbb{R}} \frac{\mu}{L(\theta - sh_n) \bar{G}(\theta - sh_n)} \left(K^{(1)} \right)^2(s) f(\theta - sh_n) ds \\ &\quad - h_n \left(\int_{\mathbb{R}} \left(K^{(1)} \right)(s) f(\theta - sh_n) ds \right)^2. \end{aligned}$$

Finally, we obtain

$$E [\Gamma_i^2] \longrightarrow \frac{\mu f(\theta)}{\bar{G}(\theta) L(\theta)} \int_{\mathbb{R}^2} \left[K^{(1)}(s) \right]^2 ds \text{ as } n \rightarrow \infty \quad (29)$$

which yields that $\Lambda_{1n} = O\left(\frac{r_n N_n}{n}\right) = O(1)$ from **B2(i)**.

Since

$$|\Lambda_{kn}| \leq \frac{2}{n} \sum_{1 \leq i < j \leq n} |cov(\Gamma_i, \Gamma_j)|, \quad k = 2, 3 \quad (30)$$

we prove that the right-hand side term in (30) tends to 0 as n tends to infinity. We use a technique developed by Masry (1986). We define the sets

$$\mathfrak{R}_1 = \{(i, j) \text{ such that } 1 \leq |i - j| \leq \rho_n\} \text{ and } \mathfrak{R}_2 = \{(i, j) \text{ such that } \rho_n + 1 \leq |i - j| \leq n - 1\}$$

where ρ_n is a sequence of integers such that $\rho_n \rightarrow \infty$, $\rho_n h_n \rightarrow 0$ as $n \rightarrow \infty$ and for some $\delta \in (0, 1)$, we have

$$h_n^{-\delta} \sum_{l=\rho_n}^{\infty} (\alpha(l))^\delta \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (31)$$

We can write

$$\frac{1}{n} \sum_{1 \leq i < j \leq n} |\text{cov}(\Gamma_i, \Gamma_j)| = \frac{1}{n} \mathfrak{S}_{1,n} + \frac{1}{n} \mathfrak{S}_{2,n} \quad (32)$$

where $\mathfrak{S}_{1,n}$ and $\mathfrak{S}_{2,n}$ be the sums of covariances over $\mathfrak{R}_1$ and $\mathfrak{R}_2$ respectively.

Furthermore, for $i < j$, we observe that

$$\begin{aligned} \text{cov}(\Gamma_i, \Gamma_j) &= E(\Gamma_i \Gamma_j) \\ &= \frac{1}{h_n} \int \int K^{(1)}\left(\frac{\theta - r}{h_n}\right) K^{(1)}\left(\frac{\theta - s}{h_n}\right) \\ &\quad f_{1,j}(r, s) dr ds \\ &\quad - \frac{1}{h_n} \int K^{(1)}\left(\frac{\theta - r}{h_n}\right) f(r) dr \\ &\quad \times \int K^{(1)}\left(\frac{\theta - s}{h_n}\right) f(s) ds. \end{aligned}$$

Then

$$\begin{aligned} |\text{cov}(\Gamma_i, \Gamma_j)| &\leq h_n \int \int K^{(1)}(m_1) K^{(1)}(m_2) \\ &\quad \times |f_{1,j}(\theta - h_n m_1, \theta - h_n m_2) \\ &\quad - f(\theta - h_n m_1) f(\theta - h_n m_2)| dm_1 dm_2. \end{aligned}$$

Using assumption **A5**, it follows that

$$|\text{cov}(\Gamma_i, \Gamma_j)| = O(h_n).$$

Hence

$$\frac{1}{n} \mathfrak{S}_{1,n} = O(h_n \rho_n). \quad (33)$$

For the second term of (32), let us use a version of the moment inequality due to Rio (1994, p. 588–590). Let p, q, γ be integer numbers such that $\frac{1}{p} + \frac{1}{q} + \frac{1}{\gamma} = 1$, it follows that

$$\frac{1}{n} \mathfrak{S}_{2,n} \leq \frac{c}{n} \sum_{(i,j) \in \mathfrak{R}_2} 2^{1+\frac{1}{\gamma}} (\alpha(|i-j|))^{\frac{1}{\gamma}} (E|\Gamma_i|^p)^{\frac{1}{p}} (E|\Gamma_i|^q)^{\frac{1}{q}}.$$

Moreover, under **A1**, we get for n large enough,

$$\begin{aligned} E|\Gamma_i|^p &\leq \frac{2^{p-1}}{h_n^{p/2}} E \left[\left| \frac{\mu \delta_i}{L(Z_i) \overline{G}(Z_i)} K^{(1)}\left(\frac{\theta - Z_i}{h_n}\right) \right|^p \right] \\ &\leq \frac{2^{p-1} \mu^{p-1} h_n}{h_n^{p/2} L^{p-1}(a) \overline{G}^{p-1}(b)} \int_{\mathbb{R}} \left| K^{(1)}(s) \right|^p f(\theta - s h_n) ds \\ &= O(h_n^{1-p/2}). \end{aligned} \quad (34)$$

Hence, using (34), it follows that

$$\begin{aligned}
 \frac{1}{n} \mathfrak{S}_{2,n} &\leq \frac{c}{n} \sum_{(i,j) \in \mathfrak{R}_2} 2^{1+\frac{1}{\gamma}} (\alpha(|i-j|))^{\frac{1}{\gamma}} \left(h_n^{1-p/2}\right)^{\frac{1}{p}} \left(h_n^{1-q/2}\right)^{\frac{1}{q}} \\
 &\leq c \left(h_n^{1-p/2}\right)^{\frac{1}{p}} \left(h_n^{1-q/2}\right)^{\frac{1}{q}} \sum_{l=\rho_n}^{\infty} (\alpha(l))^{\frac{1}{\gamma}} \\
 &\leq ch_n^{-\frac{1}{\gamma}} \sum_{l=\rho_n}^{\infty} (\alpha(l))^{\frac{1}{\gamma}}
 \end{aligned} \tag{35}$$

Therefore, by (31), we get

$$\frac{1}{n} \mathfrak{S}_{2,n} \rightarrow 0 \text{ as } n \rightarrow \infty. \tag{36}$$

Thus, from (33) and (36), we obtain

$$\frac{1}{n} \sum_{1 \leq i < j \leq n} |\text{cov}(\Gamma_i, \Gamma_j)| \rightarrow 0 \text{ as } n \rightarrow \infty. \tag{37}$$

As to $n^{-1}E \left[T_{3,n}^2 \right]$, from (29), we have

$$\begin{aligned}
 n^{-1}E \left[T_{3,n}^2 \right] &= n^{-1}E \left[\left(\sum_{i=(M_n+N_n)r_n+1}^n \Gamma_i \right)^2 \right] \\
 &\leq n^{-1} \sum_{i=(M_n+N_n)r_n+1}^n E \left[\Gamma_i^2 \right] + 2n^{-1} \sum_{1 \leq i < j \leq n} |\text{cov}(\Gamma_i, \Gamma_j)|.
 \end{aligned}$$

Therefore, (37) and assumption **B2** give the result. $\square$

Lemma 4.8. Under assumptions **A2-A5** and **B2**, we have

$$\frac{1}{n} \text{Var}(T_{1,n}) \longrightarrow \frac{\mu f(\theta)}{\overline{G}(\theta) L(\theta)} \int_{\mathbb{R}} \left[K^{(1)}(r) \right]^2 dr, \text{ as } n \rightarrow \infty.$$

Proof. We have

$$\begin{aligned}
 \frac{1}{n} \text{Var}(T_{1,n}) &= \frac{1}{n} \left(\sum_{m=1}^{r_n} \text{Var}(L_m) \right) + \frac{2}{n} \sum_{1 \leq i < j \leq r_n} \text{cov}(L_i, L_j) \\
 &\leq \frac{1}{n} \sum_{m=1}^{r_n} \sum_{i=(m-1)(M_n+N_n)+1}^{(m-1)(M_n+N_n)+M_n} E \left[\Gamma_i^2 \right] + \frac{4}{n} \sum_{1 \leq i < j \leq n} \text{cov}(\Gamma_i, \Gamma_j).
 \end{aligned}$$

Since $\frac{r_n M_n}{n} \rightarrow 1$ and from (29) and (37) the result can be achieved. $\square$

Lemma 4.9. Under the assumptions of **Lemma 4.6**, we have

$$\frac{1}{\sqrt{n}} T_{1,n} \xrightarrow{D} N(0, \sigma^2).$$

Proof. The proof is based on two steps. Firstly, it will be shown that the characteristic functions of $\frac{1}{\sqrt{n}}T_{1,n}$ minus the product of the characteristic functions of L_m , $m = 1, \dots, r_n$ converges to 0. Secondly, it is proved that the distribution determined by the product of the characteristic function of L_m , $m = 1, \dots, r_n$ is asymptotically, the distribution $N(0, \sigma^2)$.

For the first step, according to the Lemma of Volonskii and Rosanov (see Appendix in Liang and De Una-Alvarez, 2010). Note that L_m is $F_{i_m}^{j_m}$ measurable with $i_m = (m-1)(M_n + N_n) + 1$ and $j_m = (m-1)(M_n + N_n) + M_n$. Then we have

$$\left| E \left[\prod_{m=1}^{r_n} \exp \left(i t n^{-\frac{1}{2}} L_m \right) \right] - \prod_{m=1}^{r_n} E \left[\exp \left(i t n^{-\frac{1}{2}} L_m \right) \right] \right| \leq 16 r_n \alpha (N_n + 1)$$

which tends to zero by **B2(iii)**.

Now we have to show that the standard Lindeberg-Feller conditions (see Loève, 1963, p. 280) for asymptotic normality of $\frac{1}{\sqrt{n}}T_{1,n}$ under independence are satisfied. So, we have to establish

$$\frac{1}{n} \sum_{m=1}^{r_n} E \left[L_m^2 I(|L_m| > \varepsilon \sigma \sqrt{n}) \right] \rightarrow 0.$$

It suffices to show that, for every $\varepsilon > 0$, the set $\{|L_m| > \varepsilon \sigma \sqrt{n}\}$ is empty.

Since

$$\frac{1}{\sqrt{n}} |L_m| \leq \frac{M_n c}{\sqrt{n h_n}}$$

which leads to the fact that the set $\{|L_m| > \varepsilon \sigma \sqrt{n}\}$ is an empty set under assumption **B2(ii)**. Then this completes the proof of **Lemma 4.9**. $\square$

Now, to prove that $\hat{f}_n^{(2)}(\bar{\theta}_n) \rightarrow f^{(2)}(\theta)$, we have the following:

Lemma 4.10. Under assumptions **A1(i)** and **A2-A5**, we have

$$\sup_{t \in D} \left| \hat{f}_n^{(2)}(t) - f^{(2)}(t) \right| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Proof. Since

$$\begin{aligned} \left| \hat{f}_n^{(2)}(t) - f^{(2)}(t) \right| &\leq \left| \hat{f}_n^{(2)}(t) - \tilde{f}_n^{(2)}(t) \right| + \left| E \left[\tilde{f}_n^{(2)}(t) \right] - f^{(2)}(t) \right| + \left| \tilde{f}_n^{(2)}(t) - E \left[\tilde{f}_n^{(2)}(t) \right] \right| \\ &= \gamma_{1,n}(t) + \gamma_{2,n}(t) + \gamma_{3,n}(t) \end{aligned}$$

such that

$$\sup_{t \in D} \gamma_{1,n}(t) \rightarrow 0 \text{ as } n \rightarrow \infty, \quad (38)$$

$$\sup_{t \in D} \gamma_{2,n}(t) \rightarrow 0 \text{ as } n \rightarrow \infty, \quad (39)$$

$$\sup_{t \in D} \gamma_{3,n}(t) \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (40)$$

Using the same arguments as in the proof of **Lemma 4.2**, we get (38).

Since integrating by parts twice and using a change of variable, as followed in **Lemma 4.1**, we obtain

$$\begin{aligned} \sup_{t \in D} \gamma_{2,n}(t) &= \sup_{t \in D} \left| \int_{\mathbb{R}} K(r) f^{(2)}(t - rh_n) dr - f^{(2)}(t) \right| \\ &= O(h_n^2) \end{aligned}$$

we get (39). To prove (40), we use the same framework as in the proof of **Lemma 4.3**, and a suitable covering of the compact set D and changing the variables $\Delta_i(t)$ into the following

$$\begin{aligned} \Pi_i(t) &= \frac{\mu}{nh_n^3} \left[\frac{\delta_i}{L(Z_i) \bar{G}(Z_i)} K^{(2)}\left(\frac{t - Z_i}{h_n}\right) - E \left[\frac{\delta_i}{L(Z_i) \bar{G}(Z_i)} K^{(2)}\left(\frac{t - Z_i}{h_n}\right) \right] \right] \\ &= (nh_n^3)^{-1} V_i(t) \end{aligned}$$

and if the bandwidth h_n satisfies:

$$\lim_{n \rightarrow \infty} \frac{nh_n^{3(v+6)/(v-6)}}{(\log n)^{\frac{v+1}{v-6}} (\log \log n)^{\frac{6}{v-6}}} \rightarrow \infty$$

we get the result. $\square$

We observe that we have, for n large enough

$$\left| \hat{f}_n^{(2)}(\bar{\theta}_n) - f^{(2)}(\theta) \right| \leq \sup_{t \in D} \left| \hat{f}_n^{(2)}(t) - f^{(2)}(t) \right| + \left| f^{(2)}(\bar{\theta}_n) - f^{(2)}(\theta) \right|.$$

By **Lemma 4.10** and assumption **A3**, the proof of **Theorem 3.2** is completed. $\square$

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